



USER MANUAL

pygenoscape v.1.0

Overview

pygenoscape is a Python package for generating continuous spatial representations of genetic variation from pairwise genetic distance or dissimilarity data. The package is designed for exploratory visualization of spatial genetic variation using either aligned nucleotide sequence data or externally generated pairwise genetic distance matrices derived from modern population genomic workflows.

The workflow integrates:

- Pairwise genetic distance data
- Geographic coordinate projection
- Principal coordinate analysis (PCoA)
- Radial basis function (RBF) interpolation
- Interactive three-dimensional visualization

Unlike many existing approaches that depend on triangulation-based interpolation or fragmented GIS workflows, pygenoscape operates directly on full pairwise distance matrices and produces continuous spatial surfaces within a unified command-line framework.

The package supports:

- Classical sequence-based population genetic datasets
- SNP- and RAD-seq-derived genomic distance matrices
- Custom externally generated dissimilarity matrices

pygenoscape is intended primarily as a visualization and representation framework for exploratory analysis of spatial genetic turnover rather than as a formal statistical inference method.

Installation

Requirements

- Python 3.9 or higher
- macOS, Linux, or Windows

Step 1: Verify Python Installation

Open a terminal (macOS/Linux) or Command Prompt/Powershell (Windows) and run:

```
python --version
```

or, on some systems:

```
python3 --version
```

You should see output similar to:

```
Python 3.11.4
```

If Python is not installed, download it from:

<https://www.python.org/downloads>

During installation on Windows, ensure that: “Add Python to PATH” is checked.

Step 2: (Recommended) Create a Virtual Environment

Creating a virtual environment helps avoid package conflicts.

macOS/Linux

```
python3 -m venv pygenoscape-env  
source pygenoscape-env/bin/activate
```

Windows

```
python -m venv pygenoscape-env  
pygenoscape-env\Scripts\activate
```

Step 3: Install pygenoscape

```
pip install pygenoscape
```

Step 4: Verify Installation

```
pygenoscape --help
```

If installation was successful, the command-line help menu should appear.

Workflow Overview

The core workflow consists of:

1. Importing pairwise genetic distances or aligned sequence data
2. Importing geographic coordinates
3. Transforming pairwise distances into a low-dimensional representation using principal coordinate analysis (PCoA)
4. Interpolating the leading coordinate axis across geographic space
5. Generating an interactive genetic landscape visualization.

Input Modes

pygenoscape supports two major workflows:

1. FASTA Input Mode

Users provide:

- An aligned nucleotide FASTA file
- Geographic coordinates for each sample

The software internally computes pairwise uncorrected p-distances prior to interpolation.

Example command

```
pygenoscape run \  
--fasta example1.fasta \  
--coords example1_coords.csv \  
--out landscape.html
```

2. Distance Matrix Mode

Users provide:

- A precomputed symmetric pairwise distance matrix
- Geographic coordinates

This mode is recommended for:

- SNP datasets
- RAD-seq datasets
- VCF-derived distances
- microsatellite datasets

- externally generated genetic dissimilarities.

Example command

```
pygenoscape run \
--dist example2_dist.csv \
--coords example2_coords.csv \
--out landscape.html
```

Input File Formats

FASTA File

Aligned nucleotide sequences should be provided in standard FASTA format.

```
>Sample1
ATGGGCCCGGAA
>Sample2
ATGGGCCCGGAA
>Sample3
ATGGGCCCGGAT
```

Sequences must:

- be aligned
- have equal length
- contain valid nucleotide characters

Distance Matrix File

Distance matrices should be supplied as symmetric CSV files.

Example

```
,Sample1,Sample2,Sample3
Sample1,0,0.02,0.11
Sample2,0,0.02,0.08
Sample3,0.11,0.08,0
```

Matrices should:

- be square
- be symmetric
- contain identical row and column labels

The framework is agnostic to the specific distance or dissimilarity metric used.

Coordinate File

Coordinates must be provided in CSV format.

Required columns:

- id
- longitude
- latitude

Example

```
id,longitude,latitude
Sample1,-70.10,41.20
Sample2,-70.25,41.32
Sample3,-70.41,41.15
```

Coordinate identifiers must exactly match sequence or matrix sample names.

Principal Coordinate Analysis (PCoA)

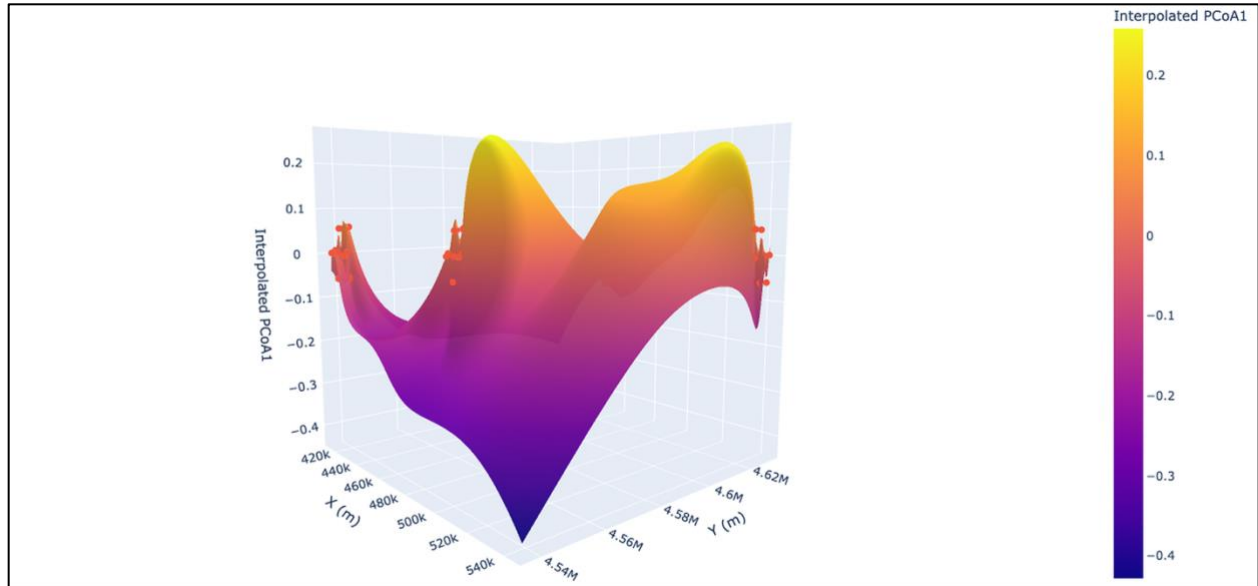
Pairwise genetic distance matrices are inherently high-dimensional and cannot be directly visualized in geographic space. pygenoscape uses principal coordinate analysis (PCoA), a form of metric multidimensional scaling, to transform pairwise distances into orthogonal coordinate axes that preserve the dominant structure present in the original data matrix. The current implementation interpolates the leading coordinate axis (PCoA1), which captures the dominant component of variation in the low-dimensional embedding. Although later axes may also contain biologically meaningful information depending on the dataset structure, PCoA1 provides a parsimonious representation of the major spatial genetic gradient.

Spatial Interpolation

The interpolated PCoA1 values are projected across geographic space using radial basis function (RBF) interpolation. Unlike triangulation-based approaches that interpolate across local spatial graphs, pygenoscape derives a low-dimensional representation from the full pairwise distance matrix prior to interpolation. This allows the framework to incorporate information from all pairwise relationships simultaneously. Sampling coordinates are automatically projected into an appropriate Universal Transverse Mercator (UTM) coordinate system to ensure interpolation occurs in metric space.

Interpreting Genetic Landscapes

The interpolated surface generated by pygenoscape represents spatial variation in interpolated PCoA1 values.



Importantly:

- Surface elevation and color encode the same interpolated variable.
- Absolute surface height should not be interpreted as a direct measure of “amount” of genetic differentiation.
- Biological interpretation should instead focus on spatial gradients and transition.

General interpretation guidelines

Smooth gradients: gradual spatial transitions typically indicate: isolation-by-distance, continuous dispersal, weak population structure.

Sharp transitions: localized steepening of gradients may indicate barriers to gene flow, reduced connectivity, phylogeographic structure, or abrupt turnover in genetic composition.

pygenoscape is intended as an exploratory visualization framework and should complement, rather than replace, formal population genetic inference methods.

Output Files

Interactive HTML Visualization

The primary output is an interactive 3-D HTML visualization that can be explored in any modern web browser. Features include rotation, zooming, panning and interactive exploration of sampling localities.

NumPy Grid Arrays

Interpolated grid values are also exported as NumPy arrays for downstream analysis or custom plotting workflows.

Command Line Parameters

Core Parameters

Parameter	Description
--fasta	Input aligned FASTA file
--dist	Input pairwise distance matrix
--coords	Coordinate CSV file
--out	Output HTML filename

Interpolation Parameters

Parameter	Description
--interp	Interpolation method (rbf or idw)
--grid-size	Resolution of interpolation grid
--height	Surface scaling behavior
--smooth	Smoothing parameter for interpolation

Example Workflow

Sequence-Based Analysis

```
pygenoscape run \  
--fasta hydroides.fasta \  
--coords hydroides_coords.csv \  
--out hydroides_landscape.html
```

SNP Distance Matrix Analysis

```
pygenoscape run \  
--dist bombus_distances.csv \  
--coords bombus_coords.csv \  
--out bombus_landscape.html
```


Computational Considerations

Pairwise distance computation scales quadratically with sample size because all pairwise comparisons must be calculated.

For large genomic datasets, external distance computation is recommended and matrix mode is generally more efficient. Interpolation runtime depends primarily on:

- number of sampling points
- grid resolution
- interpolation method

Future Development

Future versions of pygenoscape may incorporate two-dimensional contour-map visualization, interpolation of additional PCoA axes, alternative distance metrics, expanded visualization customization, and additional export formats.

Repository and Source Code

GitHub repository: <https://github.com/parasiteguy/pygenoscape>

PyPI package: <https://pypi.org/project/pygenoscape/>